دانشگاه صنعتی خواجه نصیرالدین طوسی

Research Project · Digital image processing course

CARD: Correlation Aware Restoration with Diffusion

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Presentation Date: Tir 1405 (June 2025)

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1 Abstract

This report describes a compact course implementation inspired by the paper *CARD: Correlation Aware Restoration with Diffusion*. The original paper addresses a practical weakness in many image restoration methods: they assume that image noise is independent and identically distributed Gaussian noise, while real camera sensors can produce spatially correlated noise. The project implemented here preserves the central image-processing idea of CARD by explicitly modeling correlated noise with a covariance matrix and using that covariance during restoration. The implementation is not a full diffusion or DDRM reproduction; instead, it is a reproducible CPU-friendly demonstration based on synthetic correlated noise, covariance whitening, and Wiener-style restoration.

2 Problem Statement

Many denoising and image restoration algorithms assume the measurement model

$$y = Hx_0 + n, \qquad n \sim \mathcal{N}(0, \sigma^2 I),$$

where x_0 is the clean image, H is a degradation operator, and n is i.i.d. Gaussian noise. In real imaging systems, especially rolling-shutter CMOS sensors, this assumption is often inaccurate. Sensor readout mechanisms can create row-wise and neighborhood-level noise correlations.

The correlated-noise model is:

$$y = Hx_0 + n, \qquad n \sim \mathcal{N}(0, \sigma^2 \Sigma),$$

where Σ is the covariance matrix that describes spatial noise correlation. If restoration methods ignore Σ, they may either leave structured artifacts in the output or smooth useful image details too heavily.

3 Original CARD Method

The CARD paper proposes a training-free extension of DDRM for restoration under correlated Gaussian noise. The key mathematical step is whitening. Given:

$$n \sim \mathcal{N}(0, \sigma^2 \Sigma),$$

CARD multiplies the measurement equation by $\Sigma^{-1/2}$:

$$\Sigma^{-1/2} y = \Sigma^{-1/2} H x_0 + \Sigma^{-1/2} n.$$

This produces:

$$\tilde{y} = \tilde{H} x_0 + \tilde{n}, \qquad \tilde{n} \sim \mathcal{N}(0, \sigma^2 I).$$

After this transformation, the noise satisfies the i.i.d. assumption required by DDRM-style restoration. The full paper then modifies DDRM's diffusion updates in the whitened measurement space.

4 Implemented Course Version

The implementation in this project is named `card`. It keeps the core idea of CARD but replaces the diffusion sampler with a lightweight covariance-aware restoration step. This makes the project easy to run on CPU and suitable for an image processing course demonstration.

4.1 Project Structure

File	Responsibility
`card/noise.py`	covariance model, coloring, whitening, noise generation
`card/restoration.py`	Gaussian baseline and CARD restoration
`card/metrics.py`	PSNR and SSIM metrics
`card/images.py`	synthetic image and output grid generation
`card/pipeline.py`	complete experiment pipeline
`main.py`	command-line entry point

Table 1: Main implementation files.

4.2 Pipeline

The experiment follows these steps:

1. Generate a deterministic clean synthetic image.
2. Construct a rolling-shutter-like covariance matrix.
3. Use a coloring matrix to generate correlated Gaussian noise.
4. Add this noise to the clean image.
5. Restore the noisy image using a Gaussian baseline.
6. Restore the noisy image using the CARD covariance-aware method.
7. Compute PSNR and SSIM.
8. Save a visual comparison image.

5 Covariance Model

The project uses square patches of size $p \times p$. For each pixel pair i, j inside a patch, the covariance is defined by row and column distance:

$$\Sigma_{ij} = \rho^{|r_i - r_j|}(0.45\rho)^{|c_i - c_j|} + \epsilon I,$$

where:

- r_i, c_i are the row and column coordinates of pixel i;

- ρ controls correlation strength;
- ϵI is a small regularization term.

This model creates stronger row-wise correlations than column-wise correlations, which approximates rolling-shutter banding behavior.

6 Coloring and Whitening

The covariance matrix is decomposed as:

$$\Sigma = Q\Lambda Q^T.$$

The coloring transform is:

$$C = Q\Lambda^{1/2}Q^T.$$

It maps standard Gaussian noise $z \sim \mathcal{N}(0, I)$ into correlated noise:

$$n = \sigma C z.$$

The whitening transform is:

$$W = Q\Lambda^{-1/2}Q^T.$$

It maps correlated noise back toward an i.i.d. representation:

$$W\Sigma W^T \approx I.$$

7 CARD Restoration Method

The CARD restoration function projects noisy patches into the covariance eigenbasis. In that basis, each coefficient has an estimated signal variance and noise variance. The method applies Wiener-style shrinkage:

$$g_i = \frac{\sigma^2_{\text{signal},i}}{\sigma^2_{\text{signal},i} + \sigma^2_{\text{noise},i}}.$$

The restored patch coefficients are:

$$\hat{c}_i = g_i c_i.$$

After reconstruction, the covariance-aware output is blended with a Gaussian baseline. This Gaussian result acts as a simple image prior in place of the pretrained diffusion prior used by the original CARD method.

8 How to Run

The project uses `uv` for dependency management.

```
uv sync
uv run python main.py
```

The demo writes:

```
results/card_demo.png
```

Verification commands:

```
uv run pytest
uv run ruff check .
uv run basedpyright
```

9 Results

The generated visual result is shown in Figure 1. The panels show the clean image, correlated noisy image, Gaussian baseline, and CARD-style restoration.

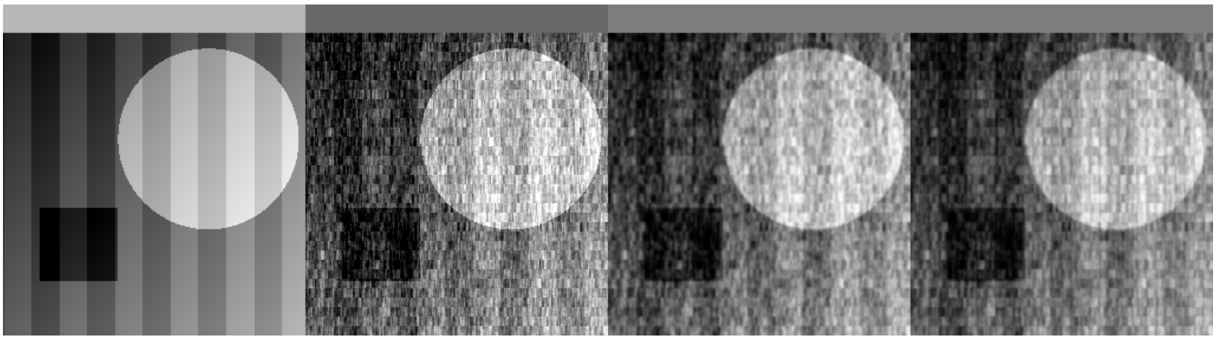

Figure 1: CARD demonstration output. Left to right: clean image, correlated noisy image, Gaussian baseline, and CARD-style restoration.

Image	PSNR	SSIM
Correlated noisy	18.62	0.889
Gaussian baseline	22.07	0.946
CARD-style restore	22.39	0.949

Table 2: Measured restoration quality in the deterministic demo.

The CARD-style method improves over both the noisy image and the Gaussian baseline in this experiment. The improvement is small but demonstrates the benefit of using the covariance structure instead of ignoring correlation.

10 Differences from the Original Paper

11 Limitations

This implementation is intentionally simplified. Its main limitations are:

- it does not implement the full DDRM diffusion sampler;
- it does not use a pretrained diffusion model;

Component	Original CARD	This Project
Restoration engine	DDRM diffusion sampler	Wiener-style denoising
Learned prior	Pretrained diffusion model	Gaussian image prior
Tasks	Denoising, deblurring, super-resolution	Denoising only
Noise data	Synthetic and real CIN-D	Synthetic only
Covariance	Dark frames or synthetic model	Hand-designed synthetic model
Evaluation	ImageNet, LSUN, CIN-D	One deterministic demo

Table 3: Comparison between the original paper and this implementation.

- it does not estimate covariance from real dark frames;
- it only demonstrates denoising;
- it evaluates on a synthetic image rather than real sensor data;
- its SSIM calculation is a simple global metric.

12 Conclusion

The CARD paper shows that explicitly modeling correlated noise can improve image restoration. This project demonstrates the same principle in a compact and reproducible form. By constructing a covariance matrix, generating correlated noise, and restoring in a covariance-aware representation, the implementation shows why the i.i.d. noise assumption can be limiting and how covariance-aware processing can improve restoration quality.